

Representation Sufficiency in Prospective Health

Query-Visible Quotients, Transport, and Enriched Response Semantics

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Paper III in a series on the formal foundations of health.
Scale-specific instantiation and measurement science of health.

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Abstract

Prospective health is evaluated by

$$\text{Health}_{\mathfrak{S},r}(x; d, t) \iff \text{Realizes}_{\mathfrak{S}}(x) \wedge \text{Adequate}_{\mathfrak{S},r}(\text{Capacity}_{\mathfrak{S}}(x; d, t)).$$

Paper I establishes the complete viable-continuation family and query-visible quotient; Paper II develops exact representation-level transport. Paper III characterizes their common representation theory.

For bundled representations

$$\text{Rep}_{u_X}(X) = \sum_{O: \text{Type}_{u_X}} (X \rightarrow O),$$

the representations sufficient for query family A are exactly the principal refinement upset above its canonical quotient:

$$\boxed{E \in \text{Suff}_A \iff E \succeq Q_A.}$$

For target family A_Q ,

$$\boxed{\text{Determinate}(R, S, q_{A_Q}) \iff \text{SuffDet}_{R,S,A_Q} \neq \emptyset,}$$

so the query quotient is the coarsest sufficient target for exact deterministic transport. For a full state-level health family H ,

$$\boxed{M \text{ sufficient for } H \iff M \succeq q_H.}$$

Control, viable cost, route structure, finite probability, and failure-aware robustness supply explicit response-grounded realizations. Scalar success probability and an attained viable minimum-cost code additionally realize complete query-language quotients by attained-range equivalence.

The core definition of health is unchanged. Lean machine checks the quotient characterizations, transport synthesis, observation criterion, compact task-visible statistics, and response-semantic boundaries.

1 Introduction

The series begins from the prospective-health normal form

$$\boxed{\text{Health}_{\mathfrak{S},r}(x; d, t) \iff \text{Realizes}_{\mathfrak{S}}(x) \wedge \text{Adequate}_{\mathfrak{S},r}(\text{Capacity}_{\mathfrak{S}}(x; d, t))}. \quad (1)$$

For state x , scenario d , and horizon t , the complete viable family is

$$\mathcal{V}_{d,t}(x) = \{(y, \gamma) \mid \gamma \in \text{Hist}(d, t, x, y), \text{Lawful}(\gamma), \text{Continues}(\gamma)\}, \quad (2)$$

and a declared observer/collector defines

$$\text{Capacity}_{\mathfrak{S}}(x; d, t) = \text{collect}\{o(v) \mid v \in \mathcal{V}_{d,t}(x)\}. \quad (3)$$

Paper I establishes these objects and the quotient retaining exactly the distinctions visible to a declared adequacy language [1]. Paper II studies how raw and health-visible representations behave under context change and state relations [2].

The state carrier and health specification are scale-neutral: the formal question applies to any organized object for which realization, continuation, prospective response, and adequacy can be scientifically specified.

Paper III asks:

$$\boxed{\begin{array}{c} \text{When does an available representation contain enough information} \\ \text{to answer a declared health-query language?} \end{array}}$$

For query family A , let q_A denote its canonical task-visible quotient. The principal criterion is

$$\boxed{\text{RepresentationSufficient}(A, e) \iff e \succeq q_A}. \quad (4)$$

The conceptual architecture separates query demand from representational supply:

$$\boxed{\begin{array}{ll} \text{declared query family } A & \longrightarrow \text{canonical task-visible information } q_A \\ \text{response semantics} & \longrightarrow \text{available representation } e. \end{array}} \quad (5)$$

Paper II supplies the complementary demand-side result: enlarging the licensed query family can refine the health-visible partition, with strictness determined by the distinctions actually consumed by the language [2]. Paper III supplies the representation-side comparison against that moving threshold and integrates it with exact deterministic transport and state-level observation. Consequently, indeterminacy of q_A from a source representation propagates to every richer A -sufficient target.

Control, viable cost, route structure, finite probability, and failure-aware robustness provide concrete response-semantic realizations of this architecture. Probability and cost additionally yield compact statistics equivalent to complete query-language quotients.

2 Inherited query-visible quotient

Fix a represented carrier X , query carrier Q , and propositional family

$$A : Q \times X \rightarrow \text{Prop}.$$

The query signature is

$$\sigma_A(x)(q) = A(q, x). \quad (6)$$

Define

$$x \sim_A y \iff \forall q \in Q, A(q, x) \leftrightarrow A(q, y). \quad (7)$$

The Paper-I quotient is

$$\boxed{q_A : X \rightarrow X/\sim_A, \quad q_A(x) = [x]_A.} \quad (8)$$

When $X = \text{Cap}$ is a capacity carrier and A is the adequacy language, write

$$\boxed{\text{HealthCap}(A) := \text{Cap}/\sim_A.} \quad (9)$$

Then $q_A : \text{Cap} \rightarrow \text{HealthCap}(A)$ is precisely the inherited health-visible capacity quotient. ¹

For an encoding

$$e : X \rightarrow O,$$

representation sufficiency is

$$\boxed{\text{RepresentationSufficient}(A, e) \iff \forall x, y, e(x) = e(y) \implies x \sim_A y.} \quad (10)$$

Paper I proves that q_A is sufficient and that every sufficient representation determines q_A uniquely on its attained range. ²

Paper III places this quotient inside a refinement order and characterizes the entire sufficient-representation class.

3 Bundled representations and the principal sufficiency theorem

3.1 A typed space of heterogeneous representations

Fix a represented carrier $X : \text{Type}_{u_X}$ and an output universe u . Define

$$\boxed{\text{Rep}_u(X) = \sum_{O : \text{Type}_u} (X \rightarrow O).} \quad (11)$$

An element is written

$$E = (O_E, e_E), \quad e_E : X \rightarrow O_E.$$

¹Formal counterpart: `lean/Tier33/Health/Paper/ResponseSemantics/RequirementSignature.lean::RequirementSignature;CapacityEquivalent;lean/Tier33/Health/Paper/ResponseSemantics/MinimalCapacityQuotient.lean::HealthCap;quotientCapacity;quotient_eq_iff.`

²Formal counterpart: `lean/Tier33/Health/Paper/ResponseSemantics/MinimalCapacityQuotient.lean::RepresentationSufficient;quotient_representation_is_sufficient;minimal_capacity_universal_property.`

Thus a representation includes both its output type and its encoding map. ³

For

$$E_1 = (O_1, e_1), \quad E_0 = (O_0, e_0),$$

define

$$E_1 \succeq E_0 \iff \forall x, y, e_1(x) = e_1(y) \implies e_0(x) = e_0(y). \quad (12)$$

This is exactly the inherited equality-kernel refinement relation applied to the bundled encodings. It is reflexive and transitive. ⁴

The unbundled refinement theorem remains useful:

$$e_1 \succeq e_0 \iff \exists! F : \text{range}(e_1) \rightarrow O_0, F(e_1(x)) = e_0(x) \quad \forall x. \quad (13)$$

⁵

3.2 Literal sufficient-representation set

Fix a query family

$$A : Q \times X \rightarrow \text{Prop}.$$

For a bundle E , define

$$E \in \text{Suff}_A \iff \text{RepresentationSufficient}(A, e_E).$$

Choose an output universe containing the canonical quotient and bundle

$$Q_A = (X/\sim_A, q_A) \in \text{Rep}_{u_X}(X). \quad (14)$$

⁶

Theorem 3.1 (Principal sufficiency theorem). *For every*

$$E \in \text{Rep}_{u_X}(X),$$

$$E \in \text{Suff}_A \iff E \succeq Q_A. \quad (15)$$

Equivalently,

$$\text{Suff}_A = \{E \in \text{Rep}_{u_X}(X) \mid E \succeq Q_A\}. \quad (16)$$

Proof. If E is sufficient, Paper I's universal property gives the unique attained-range factor from e_E to q_A , hence $E \succeq Q_A$. Conversely, if $E \succeq Q_A$, equality of E -outputs gives equality of quotient classes. Quotient equality is agreement on every declared query, which is exactly representation sufficiency. The set equality follows pointwise from this equivalence. $\square$

³Formal counterpart: `lean/Tier33/Health/Representation/Bundled.lean::BundledRepresentation;BundledRepresentation.Output;BundledRepresentation.encode;BundledRepresentation.ofFunction.`

⁴Formal counterpart: `lean/Tier33/Health/Representation/Bundled.lean::BundledRepresentation.Refines;bundled_refines_iff;bundled_refines_refl;bundled_refines_trans.`

⁵Formal counterpart: `lean/Tier33/Health/Representation/Refinement.lean::representationRefines_iff_rangeFactor.`

⁶Formal counterpart: `lean/Tier33/Health/Representation/Bundled.lean::BundledRepresentation.Sufficient;sufficientRepresentations;minimalQuotientRepresentation.`

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The output universe is fixed when forming the literal set Suff_A . Cross-universe encodings remain comparable through the inherited unbundled refinement relation. Minimality here means coarsest up to mutual equality-kernel refinement; byte size, dimension, and computational complexity are separate notions.

4 Single-query quotient and query boundaries

For a single predicate

$$P : X \rightarrow \text{Prop},$$

define

$$x \sim_P y \iff P(x) \leftrightarrow P(y). \quad (17)$$

This is an equivalence relation, giving the actual quotient

$$\boxed{q_P : X \rightarrow X/\sim_P.} \quad (18)$$

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Theorem 4.1 (Single-query principal sufficiency). *For every representation $e : X \rightarrow O$,*

$$\boxed{\text{PredicateSufficient}(e, P) \iff e \succeq q_P.} \quad (19)$$

Proof. Equality of q_P values is equivalent to agreement on P . Therefore a representation is P -sufficient exactly when every equality fiber of e lies inside one q_P -class, which is precisely $e \succeq q_P$. $\square$

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Bundling the predicate quotient as

$$Q_P = (X/\sim_P, q_P) \in \text{Rep}_{u_X}(X),$$

gives the corresponding bundled statement:

$$\boxed{E \text{ is } P\text{-sufficient} \iff E \succeq Q_P.} \quad (20)$$

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⁷Formal counterpart:

`lean/Tier33/Health/Representation/Bundled.lean::bundled_principal_upset_characterization;sufficientRepresentations_eq_principal_upset;lean/Tier33/Health/Representation/PrincipalSufficiency.lean::representationSufficient_iff_refines_minimalQuotient.`

⁸Formal counterpart: `lean/Tier33/Health/Representation/PredicateQuotient.lean::predicateSetoid;PredicateQuotient;predicateQuotient;predicateQuotient_eq_iff.`

⁹Formal counterpart: `lean/Tier33/Health/Representation/PredicateQuotient.lean::predicateQuotient_sufficient;predicateSufficient_iff_refines_predicateQuotient;predicate_principal_upset_characterization;representationEquivalent_predicateQuotient_iff.`

¹⁰Formal counterpart: `lean/Tier33/Health/Representation/Bundled.lean::predicateQuotientRepresentation;bundled_predicate_principal_upset.`

For ordinary encoding maps

$$e_0 : X \rightarrow O_0, \quad e_1 : X \rightarrow O_1,$$

a query-representation boundary

$$\text{QueryBoundary}_P(e_0, e_1)$$

contains

$$e_1 \succeq e_0, \quad \neg \text{PredicateSufficient}(e_0, P), \quad \text{PredicateSufficient}(e_1, P).$$

Theorem 4.1 strengthens its interpretation:

$$\boxed{e_0 \not\preceq q_P, \quad e_1 \succeq q_P.} \tag{21}$$

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Thus a strict boundary crosses the canonical sufficiency threshold. The rich side may retain substantial response detail beyond the minimal quotient q_P .

5 Representation classes and attained ranges

Three representation roles recur in the applications.

A *response-grounded representation* retains domain response structure such as controlled histories, viable costs, route identities, probability masses, or positive/negative response profiles.

A *sufficient representation* lies in

$$\text{Suff}_A,$$

equivalently above the canonical quotient in the refinement order.

A *minimal task-visible representation* is the canonical quotient itself, or a concrete statistic separately proved mutually refining with it.

5.1 Mutual refinement and attained-range equivalence

For

$$e_0 : X \rightarrow O_0, \quad e_1 : X \rightarrow O_1,$$

mutual refinement

$$e_0 \equiv e_1$$

induces canonical maps on attained ranges in both directions:

$$\boxed{e_0 \equiv e_1 \implies \text{range}(e_0) \simeq \text{range}(e_1).} \tag{22}$$

The equivalence sends the attained point $e_0(x)$ to the attained point $e_1(x)$. ¹²

¹¹Formal counterpart:

`lean/Tier33/Health/Representation/PredicateBoundary.lean::boundary_coarse_not_refines_predicateQuotient; boundary_rich_refines_predicateQuotient; query_boundary_crosses_predicateQuotient.`

¹²Formal counterpart:

`lean/Tier33/Health/Representation/RangeEquivalence.lean::rangeMapOfRefines; rangeMapOfRefines_commut es; rangeEquiv_of_representationEquivalent; rangeEquiv_of_representationEquivalent_commut es.`

Because q_A is surjective,

$$e \equiv q_A \implies \text{range}(e) \simeq X/\sim_A. \quad (23)$$

¹³

Thus a concrete statistic *realizes the canonical quotient* when its attained values are canonically equivalent to the query-visible quotient classes.

Coordinate enrichment supplies a different construction. For

$$e : X \rightarrow O, \quad z : X \rightarrow Z,$$

define

$$(e \oplus z)(x) = (e(x), z(x)).$$

Then

$$e \oplus z \succeq e,$$

and sufficiency is preserved. If $e(x) = e(y)$ and $z(x) \neq z(y)$, the enrichment is strict. ¹⁴

6 Deterministic transport in the representation order

6.1 Determinacy

Let

$$R : X \rightsquigarrow Y$$

be a relation, with source and target representations

$$S : X \rightarrow A, \quad T : Y \rightarrow B.$$

Paper II's determinacy condition is

$$\text{Determinate}(R, S, T) \iff \forall x, x', y, y', R(x, y) \wedge R(x', y') \wedge S(x) = S(x') \implies T(y) = T(y'). \quad (24)$$

¹⁵

When Equation (24) holds, Paper II constructs the unique deterministic factor on attained source observations. ¹⁶

¹³Formal counterpart: `lean/Tier33/Health/Representation/RangeEquivalence.lean::surjectiveRangeEquiv;rangeEquiv_minimalQuotient_of_equivalent;rangeEquiv_minimalQuotient_commutates.`

¹⁴Formal counterpart:
`lean/Tier33/Health/Representation/Refinement.lean::enrichRepresentation;enrichRepresentation_refine
s_base;enrichment_preserves_sufficiency;enrichment_strict_of_added_coordinate_split.`

¹⁵Formal counterpart:
`lean/Tier30/Research/RelationalAdmissibility/Realization/AttainedObservationTransport.lean::ObservationDeterminate.`

¹⁶Formal counterpart:
`lean/Tier30/Research/RelationalAdmissibility/Realization/AttainedObservationTransport.lean::attainedObservationTransport_existsUnique_iff.`

6.2 Source and target monotonicity

Theorem 6.1 (Source refinement monotonicity). *If*

$$S_1 \succeq S_0$$

and

$$\text{Determinate}(R, S_0, T),$$

then

$$\boxed{\text{Determinate}(R, S_1, T)}. \tag{25}$$

Proof. For two related pairs with

$$S_1(x) = S_1(x'),$$

source refinement gives

$$S_0(x) = S_0(x').$$

Apply the assumed determinacy. □

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Theorem 6.2 (Target coarsening monotonicity). *If*

$$T_1 \succeq T_0$$

and

$$\text{Determinate}(R, S, T_1),$$

then

$$\boxed{\text{Determinate}(R, S, T_0)}. \tag{26}$$

Proof. The finer target determinacy gives

$$T_1(y) = T_1(y').$$

Target refinement converts this to

$$T_0(y) = T_0(y').$$

□

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Equations (25)–(26) expose the transport-information tradeoff. Finer source information can separate source collisions. Finer target information imposes more distinctions that must remain determined.

¹⁷Formal counterpart:

`lean/Tier33/Health/Representation/Transport.lean::observationDeterminate_of_sourceRefines.`

¹⁸Formal counterpart:

`lean/Tier33/Health/Representation/Transport.lean::observationDeterminate_of_targetCoarsening.`

7 The least sufficient target for exact deterministic transport

Let

$$R : X \rightsquigarrow Y$$

be a relation, let

$$S : X \rightarrow O_S$$

be the source representation, and let

$$A_Q : Q \times Y \rightarrow \text{Prop}$$

be the target query family. Bundle target representations in

$$\text{Rep}_{u_Y}(Y).$$

Define the set of sufficient determinate targets

$$\boxed{\text{SuffDet}_{R,S,A_Q} = \left\{ E \in \text{Rep}_{u_Y}(Y) \mid \text{RepresentationSufficient}(A_Q, e_E) \wedge \text{Determinate}(R, S, e_E) \right\}.}$$
(27)

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Theorem 7.1 (Typed minimal sufficient target transport).

$$\boxed{\text{Determinate}(R, S, q_{A_Q}) \iff \text{SuffDet}_{R,S,A_Q} \neq \emptyset.}$$
(28)

Proof. Expanding Equation (27), nonemptiness is exactly the existence of a bundled A_Q -sufficient determinate target. If q_{A_Q} is determinate, choose the canonical bundle

$$Q_{A_Q} = (Y / \sim_{A_Q}, q_{A_Q}),$$

which is sufficient by construction. Conversely, choose

$$E \in \text{SuffDet}_{R,S,A_Q}.$$

The principal theorem gives

$$e_E \succeq q_{A_Q},$$

and target coarsening transports determinacy of e_E to determinacy of q_{A_Q} . □

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The original dependent target package remains available internally; explicit bridge maps connect it to the bundled formulation. ²¹

¹⁹Formal counterpart: `lean/Tier33/Health/Representation/BundledTransport.lean::sufficientDeterminateTargets;BundledSufficientDeterminateTarget.`

²⁰Formal counterpart:
`lean/Tier33/Health/Representation/BundledTransport.lean::minimalQuotient_determinate_iff_exists_bundled_sufficient_target;minimalQuotient_determinate_iff_bundled_target_nonempty.`

²¹Formal counterpart:
`lean/Tier33/Health/Representation/BundledTransport.lean::sufficientDeterminateTarget_toBundled;bundledTarget_toSufficientDeterminateTarget;sufficientDeterminateTarget_nonempty_iff_bundled_exists.`

Corollary 7.2 (Least sufficient deterministic target). *For any target encoding $e : Y \rightarrow O$,*

$$\boxed{\text{RepresentationSufficient}(A_Q, e) \wedge \text{Determinate}(R, S, e) \implies e \succeq q_{A_Q} \wedge \text{Determinate}(R, S, q_{A_Q}).} \quad (29)$$

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Thus q_{A_Q} is the least demanding sufficient target for exact deterministic transport in the equality-kernel refinement order. The phrase refers only to the number of distinctions that determinacy must preserve.

The corresponding obstruction is literal:

$$\boxed{\neg \text{Determinate}(R, S, q_{A_Q}) \iff \text{SuffDet}_{R, S, A_Q} = \emptyset.} \quad (30)$$

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A sufficient target in a larger output universe can still be compared by the unbundled theorem: determinacy followed by target coarsening produces the canonical same-universe quotient witness. The theorem is formulated for exact deterministic transport in the equality-kernel refinement order.

8 Canonical Paper-II transport witness

Paper II supplies an exact finite example of the transport-information tradeoff. In the adult/mild/now fixture,

$$R_{\text{raw}} = \{(P, B), (P, R)\}.$$

The target raw capacities are

$$C(B) = \{\text{stable}, \text{exposed}\}, \quad C(R) = \{\text{stable}\},$$

so the same source state P reaches unequal raw target observations. Raw capacity determinacy therefore fails.

Let A_{adult} be the exact licensed adult adequacy family already used by Paper II. The target health-class observation is definitionally

$$\boxed{\text{classOf}(x) = q_{A_{\text{adult}}}(C(x)).} \quad (31)$$

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The adult licensed language maps the raw-distinct baseline and recovered capacities to the same class because every licensed requirement agrees on them. Hence

$$q_{A_{\text{adult}}}(C(B)) = q_{A_{\text{adult}}}(C(R)).$$

²²Formal counterpart: `lean/Tier33/Health/Representation/BundledTransport.lean::sufficient_determinate_target_refines_and_determines_minimalQuotient.`

²³Formal counterpart: `lean/Tier33/Health/Representation/BundledTransport.lean::minimalQuotient_indeterminate_iff_bundled_targets_empty.`

²⁴Formal counterpart: `lean/Tier33/Health/PaperIII/TransportBridge.lean::adultTargetLanguage;paperII_classOf_is_queryQuotient.`

Keep the source representation fixed at raw capacity C . The only change is target coarsening:

$$C \rightsquigarrow q_{A_{\text{adult}}} \circ C.$$

Since both edges of R_{raw} have the same source state P ,

$$\boxed{\neg \text{Determinate}(R_{\text{raw}}, C, C), \quad \text{Determinate}(R_{\text{raw}}, C, q_{A_{\text{adult}}} \circ C).} \quad (32)$$

The first statement is Paper II’s raw-capacity obstruction. For the positive statement, Equation (31) gives the class/quotient identity. Hence raw capacity refines the class observation:

$$C \succeq \text{classOf},$$

because equality of raw capacities implies equality after composition with $q_{A_{\text{adult}}}$. Paper II already proves

$$\text{Determinate}(R_{\text{raw}}, \text{classOf}, \text{classOf}).$$

Applying source-refinement monotonicity to

$$C \succeq \text{classOf}$$

yields

$$\text{Determinate}(R_{\text{raw}}, C, \text{classOf}),$$

and the class/quotient identity gives the displayed target

$$q_{A_{\text{adult}}} \circ C.$$

Thus the fixed-source positive statement is an explicit corollary of the machine-checked source-refinement theorem and the existing Paper-II class-determinacy theorem. ²⁵

The existing Paper-II class observation remains useful as a state-level representation. Pull the same adult language back to states:

$$H_{\text{adult}}(r, x) = A_{\text{adult}}(r, C(x)).$$

The Paper-II class observation and the state quotient $q_{H_{\text{adult}}}$ mutually refine each other:

$$\boxed{\text{classOf} \equiv q_{H_{\text{adult}}}.} \quad (33)$$

²⁶

Thus the exact Paper-II fixture first demonstrates target coarsening with the source representation held fixed. The additional state-level equivalence result then instantiates Theorem 7.1 with a sufficient determinate target. ²⁷

The lesson is precise. Raw capacity preserves distinctions outside the licensed query language. Those additional distinctions can obstruct deterministic transport even when the canonical sufficient target transports cleanly.

²⁵ *Formal counterpart:* `lean/Tier33/Health/Development/RelationalControls.lean::rawBranching;raw_branching_obstructs_capacity_transport;raw_indeterminacy_can_be_health_determinate;lean/Tier33/Health/PaperIII/TransportBridge.lean::adultTargetLanguage;paperII_classOf_is_queryQuotient;lean/Tier33/Health/Representation/Transport.lean::observationDeterminate_of_sourceRefines.`

²⁶ *Formal counterpart:* `lean/Tier33/Health/PaperIII/TransportBridge.lean::adultStateLanguage;adult_classObservation_equivalent_stateQuotient.`

²⁷ *Formal counterpart:* `lean/Tier33/Health/PaperIII/TransportBridge.lean::paperII_fixture_has_sufficient_determinate_target.`

9 State-level health observation and the normal form

Let

$$H : Q \times X \rightarrow \text{Prop}$$

be a full state-level health family. Define

$$x \sim_H y \iff \forall q \in Q, H(q, x) \leftrightarrow H(q, y)$$

and the quotient

$$\boxed{q_H : X \rightarrow X/\sim_H.} \quad (34)$$

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For an abstract observation

$$M : X \rightarrow Y,$$

logical health-observation sufficiency is the ordinary family-sufficiency condition.

Theorem 9.1 (Health-observation sufficiency).

$$\boxed{M \text{ is sufficient for } H \iff M \succeq q_H.} \quad (35)$$

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9.1 The immutable Health normal form as a sufficient state representation

Fix a health specification, scenario d , and horizon t . Let

$$A_r(c) = \text{Adequate}_r(c), \quad C(x) = \text{Capacity}(x; d, t),$$

and define the full state family

$$\boxed{H_r(x) = \text{Realizes}(x) \wedge A_r(C(x)).} \quad (36)$$

This is the inherited health normal form, the inherited definition. ³⁰

Now define

$$\boxed{E_H(x) = (\text{Realizes}(x), q_A(C(x))) \in \text{Prop} \times (\text{Cap}/\sim_A) = \text{Prop} \times \text{HealthCap}(A).} \quad (37)$$

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²⁸Formal counterpart: `lean/Tier33/Health/Representation/HealthObservation.lean::HealthFamilyQuotient;healthFamilyQuotient;HealthObservationSufficient.`

²⁹Formal counterpart: `lean/Tier33/Health/Representation/HealthObservation.lean::healthObservationSufficient_iff_refines_healthFamilyQuotient.`

³⁰Formal counterpart: `lean/Tier33/Health/Representation/HealthNormalForm.lean::specHealthFamily.`

³¹Formal counterpart: `lean/Tier33/Health/Representation/HealthNormalForm.lean::healthNormalFormRepresentation.`

Remark 9.2. The first coordinate of E_H is proposition-valued. Equality in that coordinate is extensional under propositional extensionality, equivalently agreement of realization truth. The axiom audit records the standard use of `propext`.

The adequacy predicate descends to the capacity quotient by

$$\boxed{\bar{A}_r([c]_A) \iff A_r(c).} \quad (38)$$

This is well-defined because

$$c \sim_A c' \implies A_r(c) \leftrightarrow A_r(c').$$

For query r , define the factor

$$\boxed{F_r(L, [c]_A) = L \wedge \bar{A}_r([c]_A).} \quad (39)$$

Then

$$\boxed{F_r(E_H(x)) \iff H_r(x).} \quad (40)$$

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Consequently,

$$\boxed{\text{RepresentationSufficient}(H, E_H), \quad E_H \succeq q_H.} \quad (41)$$

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This connects the generic state-level quotient directly to the health definition: present realization and the capacity-level query quotient together form a sufficient representation for the complete health family. The theorem establishes the refinement direction $E_H \succeq q_H$.

9.2 Observation obstruction

If

$$M(x_0) = M(x_1)$$

and for some query q ,

$$\neg(H(q, x_0) \leftrightarrow H(q, x_1)),$$

then M is insufficient. Every deterministic postprocessing of the same observation remains insufficient:

$$\boxed{\neg \text{HealthObservationSufficient}(H, h \circ M) \quad \forall h : Y \rightarrow Z.} \quad (42)$$

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Equation (35) is a logical information requirement.

³²Formal counterpart: `lean/Tier33/Health/Representation/HealthNormalForm.lean::healthNormalFormFactor;specHealth_factors_through_normalFormRepresentation.`

³³Formal counterpart: `lean/Tier33/Health/Representation/HealthNormalForm.lean::healthNormalFormRepresentation_refines_healthFamilyQuotient.`

³⁴Formal counterpart: `lean/Tier33/Health/Representation/HealthObservation.lean::healthObservation_insufficient_of_query_split;postprocessing_cannot_repair_healthObservation_split.`

10 Concrete minimal task-visible statistics

The quotient q_A is abstract. In some domains a familiar statistic can be proved mutually refining with it. The attained-range equivalence theorem then identifies the statistic’s realized values with the query-visible quotient classes.

10.1 Bounded probability-threshold language

Let $\mathcal{R}$ be a finite response carrier and define the normalized rational response-mass carrier

$$\mathcal{M}_{\mathcal{R}} = \left\{ \mu : \mathcal{R} \rightarrow \mathbb{Q} \mid (\forall r \in \mathcal{R}, 0 \leq \mu(r)) \wedge \sum_{r \in \mathcal{R}} \mu(r) = 1 \right\}. \quad (43)$$

Fix an outcome predicate $P : \mathcal{R} \rightarrow \text{Prop}$. The success-probability representation is

$$p_{\text{succ}}(\cdot, P) : \mathcal{M}_{\mathcal{R}} \rightarrow \mathbb{Q}, \quad p(\mu) = p_{\text{succ}}(\mu, P).$$

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The threshold carrier is the actual subtype

$$\Theta = \{\theta \in \mathbb{Q} \mid 0 \leq \theta \leq 1\}. \quad (44)$$

The complete bounded probability-threshold family has type

$$A_{\Theta} : \Theta \times \mathcal{M}_{\mathcal{R}} \rightarrow \text{Prop}, \quad A_{\Theta}(\theta, \mu) \iff \theta \leq p(\mu). \quad (45)$$

For readability write $A_{\theta}(\mu) = A_{\Theta}(\theta, \mu)$. 36

Theorem 10.1 (Bounded-threshold profile completeness). *For any two finite response masses,*

$$[\forall \theta \in \Theta, A_{\theta}(\mu_1) \leftrightarrow A_{\theta}(\mu_2)] \iff p(\mu_1) = p(\mu_2). \quad (46)$$

Proof. Success probability lies in $[0, 1]$, so $p(\mu_1)$ and $p(\mu_2)$ themselves are valid elements of Θ . Evaluate the common threshold profile at $\theta = p(\mu_1)$ to obtain

$$p(\mu_1) \leq p(\mu_2),$$

and at $\theta = p(\mu_2)$ to obtain the reverse inequality. Antisymmetry gives equality. The converse follows by rewriting equal scalar probabilities. $\square$

37

³⁵Formal counterpart:

`lean/Tier33/Health/Paper/FAQ/FiniteProbability.lean::FiniteResponseMass;successProbability.`

³⁶Formal counterpart: `lean/Tier33/Health/PaperIII/BoundedProbability.lean::`

`ProbabilityThreshold;boundedProbabilityThresholdLanguage.`

³⁷Formal counterpart: `lean/Tier33/Health/PaperIII/BoundedProbability.lean::`

`bounded_thresholds_agree_iff_successProbability_equal.`

Therefore

$$p_{\text{succ}}(\cdot, P) \equiv q_{A_\Theta}, \quad q_{A_\Theta} : \mathcal{M}_{\mathcal{R}} \rightarrow \mathcal{M}_{\mathcal{R}} / \sim_{A_\Theta}. \quad (47)$$

and this mutual refinement yields the actual attained-range identification

$$\text{range}(p_{\text{succ}}) \simeq \mathcal{M}_{\mathcal{R}} / \sim_{A_\Theta}. \quad (48)$$

The equivalence maps the attained scalar value of each distribution to that distribution's quotient class.³⁸

Scalar success probability is thus a concrete minimal task-visible realization of the bounded rational probability-threshold quotient. Positive support and the full response law remain richer representations.

10.2 All-budget cost language

Return first to the state-level budget construction. The cost object is history-owned:

$$c : \text{Hist}(d, t, x, y) \rightarrow K \quad (49)$$

³⁹ for every indexed history type $\text{Hist}(d, t, x, y)$. For source state x , budget b , and observer atom a ,

$$B_b(x) = \{a \mid \exists v \in \mathcal{V}_{d,t}(x), c(v.\gamma) \leq b \wedge o(v) = a\}. \quad (50)$$

The budget-restricted capacity is

$$\text{BudgetCapacity}_b(x) = \text{collect}(B_b(x)). \quad (51)$$

For requirement r ,

$$\text{BudgetHealth}_{r,b}(x) \iff \text{Realizes}(x) \wedge \text{Adequate}_r(\text{BudgetCapacity}_b(x)). \quad (52)$$

⁴⁰

Let K be a partially ordered cost carrier. Fix observer O , cost c , context d, t , and observed atom a . Define the genuine source subtype

$$\mathcal{D}_a = \left\{ x \mid \begin{array}{l} a \text{ is unavailable from } x \text{ through viable continuation,} \\ \text{or an attained minimal viable cost for observing } a \text{ exists} \end{array} \right\}. \quad (53)$$

⁴¹

The code is the typed function

$$\text{CostCode}_a : \mathcal{D}_a \rightarrow \text{Option } K. \quad (54)$$

³⁸Formal counterpart: `lean/Tier33/Health/PaperIII/BoundedProbability.lean::successProbability_equivalent_boundedThresholdQuotient;successProbability_rangeEquiv_boundedThresholdQuotient;successProbability_boundedRangeEquiv_commutates.`

³⁹Formal counterpart: `lean/Tier33/Health/Paper/FAQ/CostAndReserve.lean::HistoryCost.`

⁴⁰Formal counterpart:

`lean/Tier33/Health/Paper/FAQ/CostAndReserve.lean::budgetAtoms;budgetCapacity;BudgetHealth.`

⁴¹Formal counterpart: `lean/Tier33/Health/PaperIII/CostCode.lean::AtomCostCodeDomain.`

For $x \in \mathcal{D}_a$,

$$\text{CostCode}_a(x) = \text{none}$$

iff a is unavailable through every viable continuation, and

$$\text{CostCode}_a(x) = \text{some}(m)$$

iff m is the attained minimal viable observed-atom cost. ⁴²

The complete fixed-atom budget language is the family

$$A_{\text{budget}} : K \times \mathcal{D}_a \rightarrow \text{Prop}, \quad A_{\text{budget}}(b, x) \iff a \in B_b(x). \quad (55)$$

Write $A_b(x) = A_{\text{budget}}(b, x)$. Then

$$A_b(x) \iff \exists m, \text{CostCode}_a(x) = \text{some}(m) \wedge m \leq b. \quad (56)$$

⁴³

The complete budget profile determines the code:

$$\forall x, y \in \mathcal{D}_a, \quad [\forall b \in K, A_b(x) \leftrightarrow A_b(y)] \iff \text{CostCode}_a(x) = \text{CostCode}_a(y). \quad (57)$$

⁴⁴

Therefore

$$\text{CostCode}_a \equiv q_{A_{\text{budget}}}, \quad q_{A_{\text{budget}}} : \mathcal{D}_a \rightarrow \mathcal{D}_a / \sim_{A_{\text{budget}}}. \quad (58)$$

and the realized code values are canonically equivalent to the whole budget-language quotient:

$$\text{range}(\text{CostCode}_a) \simeq \mathcal{D}_a / \sim_{A_{\text{budget}}}. \quad (59)$$

⁴⁵

The finite cost fixture realizes

$$\text{CostCode}(\text{true}) = \text{some}(1), \quad \text{CostCode}(\text{false}) = \text{some}(3).$$

⁴⁶

Probability and cost therefore provide literal attained-range realizations of their complete task-visible quotients.

⁴²Formal counterpart: `lean/Tier33/Health/PaperIII/CostCode.lean::observedAtomCostCode;observedAtomCostCode_eq_some_iff;observedAtomCostCode_eq_none_iff.`

⁴³Formal counterpart: `lean/Tier33/Health/PaperIII/CostCode.lean::atomBudgetLanguage;costCodeAffordable;budgetMembership_iff_costCode.`

⁴⁴Formal counterpart: `lean/Tier33/Health/PaperIII/QuotientRealizations.lean::all_budgetQueries_agree_iff_observedAtomCostCode_equal.`

⁴⁵Formal counterpart: `lean/Tier33/Health/PaperIII/CostCode.lean::observedAtomCostCode_equivalent_budgetLanguageQuotient;lean/Tier33/Health/PaperIII/QuotientRealizations.lean::observedAtomCostCode_rangeEquiv_budgetLanguageQuotient;observedAtomCostCode_rangeEquiv_commutates.`

⁴⁶Formal counterpart: `lean/Tier33/Health/PaperIII/CostCode.lean::CostCodeFixture.source;CostCodeFixture.actual_minimum_codes.`

11 Compressed route summary

A route capacity is the typed object

$$C = (\mathcal{R}_C, o_C, \iota_C), \quad \mathcal{R}_C \subseteq \text{Route}, \quad o_C : \text{Route} \rightarrow A, \quad \iota_C : \text{Route} \rightarrow I. \quad (60)$$

Only the restrictions of o_C and ι_C to $\mathcal{R}_C$ enter the displayed queries. The endpoint-set projection is

$$\pi_{\text{set}}(C) = o_C[\mathcal{R}_C]. \quad (61)$$

47

For a finite route carrier ‘Route’, atom carrier A , and declared class carrier I , the summary has the exact type

$$\text{RouteSummary} : \text{RouteCapacity}(\text{Route}, A, I) \longrightarrow \mathcal{P}(A) \times \mathbb{N} \times \mathcal{P}(I). \quad (62)$$

Its value is

$$\text{RouteSummary}(C) = (\pi_{\text{set}}(C), |\mathcal{R}_C|, \iota_C[\mathcal{R}_C]). \quad (63)$$

48

The summary refines endpoint projection:

$$\text{RouteSummary} \succeq \pi_{\text{set}}.$$

For finite route sets,

$$\text{HasDistinctRoutes}(C) \iff 1 < |\mathcal{R}_C|. \quad (64)$$

Distinct declared classes are determined by the class image:

$$\text{HasDistinctIndependenceClasses}(C) \iff \exists i_1, i_2 \in \iota_C[\mathcal{R}_C], i_1 \neq i_2. \quad (65)$$

49

In the generated fixture the one-route and two-route systems have route counts 1 and 2. ⁵⁰

The route summary is a compressed response-grounded sufficient representation for the displayed route queries. Minimality and physical-independence semantics remain separate questions.

The generated one-route/two-route fixture has identical endpoint-set projection and distinct route multiplicity and class-image truth. Thus endpoint projection lies below the relevant query thresholds, and full route capacity together with the compressed summary retain the distinctions consumed by those queries. ⁵¹

⁴⁷ Formal counterpart:

`lean/Tier33/Health/Paper/FAQ/RouteRedundancy.lean::RouteCapacity;routeSetProjection.`

⁴⁸ Formal counterpart: `lean/Tier33/Health/PaperIII/RouteSummary.lean::RouteSummary.`

⁴⁹ Formal counterpart: `lean/Tier33/Health/PaperIII/RouteSummary.lean::routeSummary_refines_endpointProjection;distinctRoutes_iff_routeCount;distinctClasses_iff_classImage;routeSummary_sufficient_for_distinctRoutes;routeSummary_sufficient_for_distinctClasses.`

⁵⁰ Formal counterpart: `lean/Tier33/Health/PaperIII/RouteSummary.lean::generated_routeSummary_counts.`

⁵¹ Formal counterpart: `lean/Tier33/Health/PaperIII/RouteRepresentation.lean::routeMultiplicityQueryBoundary;routeIndependenceClassQueryBoundary.`

12 Response-grounded strict boundaries

The canonical quotient identifies the minimum task-visible information. Domain response models can retain additional semantics. The strict boundaries below show concrete response-grounded representations crossing their canonical query thresholds.

12.1 Control

A controlled dynamics consists of a policy carrier Π , action carrier U , action selector

$$a : \Pi \times X \rightarrow U,$$

admissibility predicate

$$\text{Adm} : X \times U \rightarrow \text{Prop},$$

and generated-history predicate

$$\text{Gen} : U \times \text{Hist}(d, t, x, y) \rightarrow \text{Prop}$$

for each indexed history type $\text{Hist}(d, t, x, y)$.⁵²

For fixed scenario d and horizon t , take a dependent outcome family

$$\boxed{P_{x,y} : \text{Hist}(d, t, x, y) \rightarrow \text{Prop}.} \quad (66)$$

Equivalently, for a fixed source x , this is a predicate on the outgoing-history sigma carrier

$$\text{Hist}_{d,t}(x) = \bigsqcup_{y:X} \text{Hist}(d, t, x, y).$$

For fixed controlled dynamics C and source x , define the fixed-carrier response set

$$\boxed{R_C(x) = \{\rho \mid \begin{aligned} &\rho.\text{source} = x \\ &\wedge \text{Adm}(\rho.\text{source}, a(\rho.\pi, \rho.\text{source})) \\ &\wedge \text{Lawful}(\rho.\gamma) \wedge \text{Gen}(a(\rho.\pi, \rho.\text{source}), \rho.\gamma) \end{aligned}\},} \quad (67)$$

where

$$\rho = (\rho.\text{source}, \rho.\text{target}, \rho.\pi, \rho.\gamma), \quad \rho.\gamma : \text{Hist}(d, t, \rho.\text{source}, \rho.\text{target}).$$

On the source- x fiber this is equivalently the set of triples

$$(y, \pi, \gamma)$$

with

$$\gamma : \text{Hist}(d, t, x, y),$$

admissibility, lawfulness, and generated-history membership.⁵³

⁵²Formal counterpart: `lean/Tier33/Health/Paper/FAQ/EndogenousControl.lean::ControlledDynamics.`

⁵³Formal counterpart: `lean/Tier33/Health/PaperIII/ControlRepresentation.lean::IndexedControlledResponse; controlledLawfulResponses.`

Endogenous recovery factors explicitly:

$$\boxed{\text{EndogenousRecovery}(C, P, x) \iff \exists \rho \in R_C(x), P_{\rho.\text{source}, \rho.\text{target}}(\rho.\gamma).} \quad (68)$$

54

For a fixed policy π , define

$$R_{C,\pi}(x) = \{\rho \in R_C(x) \mid \rho.\pi = \pi\}.$$

Then robust endogenous recovery is

$$\boxed{\text{RobustEndogenousRecovery}(C, P, x) \iff \exists \pi, R_{C,\pi}(x) \neq \emptyset \wedge \forall \rho \in R_{C,\pi}(x), P_{\rho.\text{source}, \rho.\text{target}}(\rho.\gamma).} \quad (69)$$

55

In the enabled/disabled fixture, lawful return possibility is identical at the coarse level and

$$R_{\text{disabled}} = \emptyset, \quad R_{\text{enabled}} = \{\text{recoverRecord}, \text{adaptRecord}\}.$$

The coarse representation lies below q_P for endogenous recovery; the controlled-response representation lies above it. ⁵⁶

12.2 Viable cost and budget health

Define the costed viable-response data

$$\boxed{D_c(x) = \{(o(v), c(v.\gamma)) \mid v \in \mathcal{V}(x)\}.} \quad (70)$$

For any such data set D and budget b ,

$$\boxed{B_b(D) = \{a \mid \exists k, (a, k) \in D \wedge k \leq b\}.} \quad (71)$$

The cost-data projection recovers the original budget atom set exactly:

$$\boxed{B_b(D_c(x)) = B_b(x).} \quad (72)$$

57

The rich state representation is

$$\boxed{E_{\text{cost}}(x) = (\text{Realizes}(x), D_c(x)).} \quad (73)$$

⁵⁴Formal counterpart: `lean/Tier33/Health/PaperIII/ControlRepresentation.lean::controlledRecoveryFactor;endogenousRecovery_factors_through_controlledResponses.`

⁵⁵Formal counterpart: `lean/Tier33/Health/PaperIII/ControlRepresentation.lean::robustControlledRecoveryFactor;robustEndogenousRecovery_factors_through_controlledResponses.`

⁵⁶Formal counterpart: `lean/Tier33/Health/PaperIII/ControlRepresentation.lean::ControlBoundaryFixture.disabled_responses_empty;ControlBoundaryFixture.enabled_responses_exact;controlQueryBoundary;lean/Tier33/Health/Representation/PredicateBoundary.lean::query_boundary_crosses_predicateQuotient.`

⁵⁷Formal counterpart: `lean/Tier33/Health/PaperIII/RepresentationApplications.lean::budgetAtoms_eq_costedProjection.`

For query r and budget b , define the factor

$$\boxed{F_{r,b}(L, D) = L \wedge \text{Adequate}_r(\text{collect}(B_b(D)))}. \quad (74)$$

Using Equation (72),

$$\begin{aligned} F_{r,b}(E_{\text{cost}}(x)) &= \text{Realizes}(x) \wedge \text{Adequate}_r(\text{collect}(B_b(D_c(x)))) \\ &= \text{Realizes}(x) \wedge \text{Adequate}_r(\text{BudgetCapacity}_b(x)). \end{aligned} \quad (75)$$

Therefore

$$\boxed{F_{r,b}(E_{\text{cost}}(x)) \iff \text{BudgetHealth}_{r,b}(x)}. \quad (76)$$

58

Thus E_{cost} is sufficient for fixed-budget health and refines complete uncosted capacity. The finite Boolean fixture gives the strict boundary. ⁵⁹

12.3 Probability support and mass

Logical success, positive successful support, and scalar success probability are distinct:

$$\begin{aligned} S_{\text{logic}}(P) &= \{r \mid P(r)\}, \\ S_{\mu}^+(P) &= \{r \mid P(r) \wedge 0 < \mu(r)\}, \\ p_{\text{succ}}(\mu, P) &= \sum_r \mathbf{1}_{P(r)} \mu(r). \end{aligned}$$

The high and low distributions have equal positive support and success probabilities 999/1000 and 1/1000. Thus positive support is insufficient for the 1/2 threshold query. The enriched pair

$$(S_{\mu}^+(P), p_{\text{succ}}(\mu, P))$$

is a strict sufficient refinement. ⁶⁰

The response-grounded pair contains the scalar success-probability coordinate. That scalar is sufficient for the complete bounded rational probability-threshold language, and Section 10 identifies it as the minimal task-visible quotient realization. ⁶¹

⁵⁸Formal counterpart: `lean/Tier33/Health/PaperIII/CostRepresentation.lean::budgetHealthRepresentation; budgetHealthRepresentationFactor; budgetHealth_factors_through_costedViableRepresentation.`

⁵⁹Formal counterpart: `lean/Tier33/Health/PaperIII/CostRepresentation.lean::budgetHealthRepresentation_sufficient; costedViableRepresentation_refines_completeCapacity; costBudgetQueryBoundary.`

⁶⁰Formal counterpart: `lean/Tier33/Health/PaperIII/ProbabilitySupport.lean::ProbabilityFixture.high_low_positiveSupport_equal; lean/Tier33/Health/PaperIII/ProbabilityRepresentation.lean::probabilityEnrichedRepresentation; probabilityThresholdQueryBoundary.`

⁶¹Formal counterpart: `lean/Tier33/Health/PaperIII/ProbabilityRepresentation.lean::successProbability_sufficient_for_thresholdAdequacy; lean/Tier33/Health/PaperIII/BoundedProbability.lean::successProbability_equivalent_boundedThresholdQuotient.`

12.4 Failure-aware robustness

Let $\mathcal{C}$ be the challenge carrier, T the horizon carrier, and $\mathcal{H}$ the response-history carrier. A challenge system supplies

$$\text{responds} : X \times \mathcal{C} \times T \times \mathcal{H} \rightarrow \text{Prop}.$$

Let $D \subseteq \mathcal{C}$ be the declared challenge family, let

$$\tau : \mathcal{C} \rightarrow T$$

assign its declared horizon, and fix an outcome predicate

$$P : \mathcal{H} \rightarrow \text{Prop}.$$

Then the response relation at challenge c is

$$\text{responds}(x, c, \tau(c), h), \quad h \in \mathcal{H}.$$

⁶²

Define the generated challenge profile by

$$G^{\text{adm}}(x) = \{c \in D \mid \exists h, \text{responds}(x, c, \tau(c), h)\}, \quad (77)$$

$$G^+(x) = \{c \in D \mid \exists h, \text{responds}(x, c, \tau(c), h) \wedge P(h)\}, \quad (78)$$

and

$$G^-(x) = \{c \in D \mid \exists h, \text{responds}(x, c, \tau(c), h) \wedge \neg P(h)\}. \quad (79)$$

Thus

$$G(x) = (G^{\text{adm}}(x), G^+(x), G^-(x)).$$

Classical outcome dichotomy gives

$$G^{\text{adm}}(x) = G^+(x) \cup G^-(x). \quad (80)$$

⁶³

The positive/negative pair therefore retains the generated profile information required by robust adequacy. ⁶⁴

The rich full-persistence representation is

$$E_{\text{rob}}(x) = (\text{PresentHealth}(x), G^+(x), G^-(x)). \quad (81)$$

For declared challenge set D , define

$$F_{\text{rob}}(H_0, G_+, G_-) = H_0 \wedge \forall c \in D, c \in G_+ \wedge c \notin G_-. \quad (82)$$

⁶²Formal counterpart:

`lean/Tier33/Health/Resilience/ChallengeFamily.lean::ChallengeSystem;DeclaredChallengeFamily.`

⁶³Formal counterpart: `lean/Tier33/Health/Paper/ResponseSemantics/RobustFactorization.lean::ChallengeProfile;challengeProfile;admitted_eq_success_union_failure.`

⁶⁴Formal counterpart: `lean/Tier33/Health/Paper/ResponseSemantics/ProfileReconstruction.lean::consistentChallengeProfileEquivOutcomeProfile;robustAdequate_iff_outcomeProfile.`

Inherited boundary	Paper-III closure
Paper I: positive prospective capacity leaves failure-sensitive robustness unresolved	$(\text{PresentHealth}, G^+, G^-)$ is sufficient for full robust persistence; G^+ alone is insufficient.
Paper I: present-state observation can lose prospective health distinctions	For every full health family H , M is sufficient exactly when $M \succeq q_H$; a splitting observation fiber remains insufficient under deterministic postprocessing.
Paper II: raw and health-visible determinacy can differ	Paper II exhibits raw target indeterminacy together with determinate canonical query-visible target transport through q_A .
Paper II: control, cost, route structure, probability, and failure-aware robustness were outside its transport scope	Each domain now has explicit response semantics and strict sufficient representation boundaries.
Abstract minimal information remained separate from rich response evidence	p_{succ} realizes the complete bounded rational probability-threshold quotient; CostCode_a realizes the all-budget quotient on its justified subdomain.

Table 1: Series-level closure.

Then

$$\boxed{\text{RobustPersistence}(x) \iff F_{\text{rob}}(E_{\text{rob}}(x))}. \quad (83)$$

⁶⁵

The successful-only and mixed systems have equal G^+ and opposite robust truth because the mixed system also contains a failed admitted response. Thus G^+ lies below the robust-persistence qp threshold and E_{rob} lies above it. ⁶⁶

13 Series-level closure

The representation theory closes explicit boundaries inherited from Papers I–II.

Together these closures place query-visible compression, response enrichment, and exact deterministic transport in one representation order.

⁶⁵ *Formal counterpart:* `lean/Tier33/Health/PaperIII/RobustRepresentation.lean::robustPersistenceRepresentation;robustPersistenceFactor;robustPersistence_factors_through_presentOutcomeProfile.`

⁶⁶ *Formal counterpart:* `lean/Tier33/Health/PaperIII/RobustRepresentation.lean::robustPersistenceRepresentation_sufficient;robustPersistenceQueryBoundary;lean/Tier33/Health/Representation/PredicateBoundary.lean::query_boundary_crosses_predicateQuotient.`

14 Related mathematical and health work

Action-theoretic and capability-oriented accounts of health motivate distinctions among possibility, available action, resources, and adaptive response [11, 12, 13, 10]. The present work assigns these distinctions explicit response representations and query-relative sufficiency statements.

The control and cost constructions are adjacent to reachability, viability theory, dynamic programming, and optimal control [3, 4]. Route structure is adjacent to reliability theory [5], and finite probability to probabilistic transition systems and model checking [6]. These neighboring theories supply broader domain-specific semantics; the present results isolate the representation distinctions consumed by the declared health queries.

The representation calculus is related to sufficient statistics, statistical experiment comparison, quotient semantics, and abstraction [7, 8, 9]. Fiber factorization, equality-kernel refinement, and quotient universal properties are standard mathematics. The order used here is a deterministic equality-kernel preorder, integrated into one machine-checked prospective-health representation theory.

15 Discussion

The principal result puts query demand and representational supply on the same refinement order. A query language fixes the distinctions that must remain visible; a response representation determines which distinctions are retained.

The same order exposes the transport-information tradeoff. Finer target representations preserve more distinctions and impose stronger exact determinacy requirements. Query sufficiency sets the lower admissible resolution, placing the canonical quotient at the interface between enough information for the query and enough coarsening for transport. Paper II’s raw/visible fixture gives the finite witness: raw target capacity is indeterminate, whereas the adult licensed quotient is determinate from the same source representation.

At state level, a collapsed observation fiber remains collapsed under deterministic postprocessing. The Health-normal-form representation shows how present realization and query-visible prospective capacity jointly provide the information required by the full health family.

The applications distinguish response-grounded evidence from task-visible information. Controlled responses, costed viable responses, route structure, probability laws, and challenge profiles retain domain semantics; selected query languages also admit smaller concrete statistics equivalent to their canonical quotients.

16 Scope boundaries

The theory concerns exact propositional query families, deterministic equality-kernel refinement, and exact relation-level transport. The terms “minimal,” “coarsest,” and “easiest” refer to this refinement order; code length, Euclidean dimension, entropy, computational cost, experimental burden, and statistical optimality define different comparison problems.

Domain-specific results retain the hypotheses stated in their sections. In particular, ‘CostCode’ uses the unavailable-or-attained-minimum subdomain, probability minimality is relative to the complete bounded rational probability-threshold language with fixed outcome predicate, and route-class semantics require independent scientific justification.

The formal query families specify information requirements. Biological semantics, observation design, sensing, calibration, uncertainty, statistical estimation, validity, reliability, and predictive comparison remain separate scientific obligations.

17 Conclusion

Paper III closes the exact deterministic representation-theoretic arc developed across Papers I–II:

$$\begin{array}{l}
 E \in \text{Suff}_A \iff E \succeq Q_A, \\
 \text{Determinate}(R, S, q_{A_Q}) \iff \text{SuffDet}_{R,S,A_Q} \neq \emptyset, \\
 M \text{ sufficient for } H \iff M \succeq q_H.
 \end{array} \tag{84}$$

These results identify the canonical information demanded by a query language, the least sufficient target for exact deterministic transport, and the logical information threshold for a state observation. The domain applications supply explicit response-semantic realizations. Probability and cost also furnish concrete statistics equivalent to complete query-language quotients.

Future work will develop general and specific instantiations of the framework across scales and extend the formal measurement-science treatment of observation, calibration, uncertainty, validation, reliability, and comparison.

A Formal source and verification

The formal source for this manuscript is pinned to

```
bd406eb6dc40dc51688a79885f6b1b43aedd9827.
```

The preferred Paper-III import root is

```
lean/Tier33/Health/PaperIII.lean.
```

The one-command verification gate is

```
python3 scripts/verify_t33_health_p3_exposure04.py -compile.
```

The recorded final gate audited a 6,074-source local import cut and compiled 73 scoped outputs with a clean compiler result. It resolved 351 unique planned and inherited formal references. The axiom audit covered 244 named declarations and projections and reported only the accepted standard dependencies `propext`, `Quot.sound`, and `Classical.choice` where required. The verifier preserved 172 preceding source/support owners byte-for-byte and all 85 prior completion fields and assignments, then added 12 substantive exposure propositions for a 97-field certificate.

The formal development uses Lean 4.28.0-rc1 and Mathlib revision

```
5352afccd6866369be9de43f5b7ec47203555f44.
```

The exact remaining dependency pins are recorded in `lake-manifest.json`. ‘Health.Health’ and the protected Paper-I/Paper-II theorem owners remain unchanged.

The inherited project defines an empty default Lake target, so proof coverage is supplied by the scoped verifier, which elaborates the preferred root, completion certificate, axiom audit, exact formal-reference surface, and mathematical-exposure gates.

B Theorem map

ID	Section	Mathematical statement	Formal counterpart
P3-01	Bundled representations	$\text{Rep}_u(X)$ bundles the output type and encoding; sufficient representations form a literal principal upset above Q_A .	<code>Representation/Bundled .lean::BundledRepresent ation;sufficientRepres entations_eq_principal _upset</code>
P3-02	Predicate quotient	q_P is the actual quotient by predicate agreement; E is P -sufficient exactly when the bundled representation refines Q_P .	<code>Representation/Bundled .lean::predicateQuotien tRepresentation;bundle d_predicate_principal_ upset</code>
P3-03	Range equivalence	Mutual refinement gives an actual equivalence of attained ranges; equivalence to q_A realizes the whole quotient on the attained statistic.	<code>Representation/RangeEq uivalence.lean::rangeE quiv_of_representation Equivalent;rangeEquiv_ minimalQuotient_of_equ ivalent</code>
P3-04	Transport monotonicity	Source refinement preserves determinacy; target coarsening preserves determinacy.	<code>Representation/Transpo rt.lean::observationDet erminate_of_sourceRefi nes;observationDetermi nate_of_targetCoarseni ng</code>
P3-05	Typed target existential	q_A is determinate iff the literal set of bundled sufficient determinate targets is nonempty.	<code>Representation/Bundled Transport.lean::minima lQuotient_determinate_ iff_exists_bundled_suf ficient_target;minimal Quotient_determinate_i ff_bundled_target_none empty</code>
P3-06	Least sufficient target	Any sufficient determinate target refines q_A and forces q_A determinacy.	<code>Representation/Bundled Transport.lean::suffic ient_determinate_targe t_refines_and_determin es_minimalQuotient</code>
P3-07	Paper-II bridge	With raw source representation fixed, raw target capacity is indeterminate and the adult query quotient target is determinate; the class observation is the same quotient observation.	<code>Development/Relational Controls.lean::raw_bran ching_obstructs_capaci ty_transport;raw_inde terminacy_can_be_health _determinate;PaperIII/ TransportBridge.lean:: paperII_classOf_is_que ryQuotient;Representat ion/Transport.lean::ob servationDeterminate_o f_sourceRefines</code>
P3-08	Health observation	An observation is sufficient for a full health family iff it refines q_H ; deterministic postprocessing cannot repair a query-splitting observation fiber.	<code>Representation/HealthO bservation.lean::health ObservationSufficient_ iff_refines_healthFami lyQuotient;postprocess ing_cannot_repair_heal thObservation_split</code>
P3-09	Health normal form	$E_H = (\text{Realizes}, q_A \circ C)$ explicitly factors every full Health proposition and refines the state-level health quotient.	<code>Representation/HealthN ormalForm.lean::specHe alth_factors_through_n ormalFormRepresentatio n;healthNormalFormRepr esentation_sufficient; healthNormalFormRepres entation_refines_healt hFamilyQuotient</code>

ID	Section	Mathematical statement	Formal counterpart
P3-10	Bounded probability	Agreement on every bounded rational probability threshold is equivalent to equality of success probability.	PaperIII/BoundedProbability.lean::bounded_thresholds_agree_iff_successProbability_equal
P3-11	Probability realization	Success probability is equivalent to, and its attained range is canonically equivalent to, the bounded rational probability-threshold quotient.	PaperIII/BoundedProbability.lean::successProbability_equivalent_boundedThresholdQuotient;successProbability_rangeEquiv_boundedThresholdQuotient
P3-12	Cost code	$\text{CostCode}_a : \mathcal{D}_a \rightarrow \text{Option}K$ determines all budgets; its attained range is canonically equivalent to the all-budget quotient.	PaperIII/CostCode.lean::observedAtomCostCode;observedAtomCostCode_equivalent_budgetLanguageQuotient;PaperIII/QuotientRealizations.lean::observedAtomCostCode_rangeEquiv_budgetLanguageQuotient
P3-13	Control factor	The exact admissibly generated lawful response set determines endogenous and policy-grouped robust endogenous recovery.	PaperIII/ControlRepresentation.lean::controlledLawfulResponses;endogenousRecovery_factors_through_controlledResponses;robustEndogenousRecovery_factors_through_controlledResponses
P3-14	Budget factor	The explicit $F_{r,b}$ reconstructs budget health from present realization and costed viable-response data.	PaperIII/CostRepresentation.lean::budgetHealthRepresentationFactor;budgetHealth_factors_through_costedViableRepresentation
P3-15	Route summary	$\text{RouteSummary} : \text{RouteCapacity} \rightarrow \mathcal{P}(A) \times \mathbb{N} \times \mathcal{P}(I)$ is sufficient for route-multiplicity and class-image queries.	PaperIII/RouteSummary.lean::RouteSummary;routeSummary_sufficient_for_distinctRoutes;routeSummary_sufficient_for_distinctClasses
P3-16	Robust factor	The explicit F_{rob} reconstructs full robust persistence from present health and positive/negative outcome coordinates.	PaperIII/RobustRepresentation.lean::robustPersistenceFactor;robustPersistence_factors_through_presentOutcomeProfile
P3-17	Completion	The Paper-III certificate contains the inherited theory, synthesis results, bundled surfaces, range realizations, bounded probability, and Health-normal-form bridge as substantive propositions.	PaperIII/Completion.lean::t33HealthPaperIII_complete

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